

Discrete Structures

Lecture # 5

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Predicate Logic

- Can be considered as a model to investigate certain arguments that cannot be expressed in propositional logic.
- Given the following propositions as premises
 - All human beings are mammals
 - Peter is a human being

We should be able to conclude that

- Peter is a mammal.
- This raises two questions
 - How to express propositions like the first premise?
 - How to provide rules to help judge the validity of the argument?

Predicate Logic

- In propositional logic we may state.
 - “Peter is a human being”
- and
- “Ann is a human being”
- But we have no means to express that the two propositions are about the same property: “**Is a human being**”

Predicate Logic

- A predicate $p(x)$ describes a property, say “ x ” is a human being.

where x is a free variable that may be substituted by values in the **Universe of discourse** (UOD also called domain) of the predicate.

- For Example, we can write the two propositions on the previous slide as: $p(\text{peter})$ and $p(\text{Ann})$.

Predicates

Statements involving variables, such that

“ $X > 3$ ”, “ $X = Y + 3$ ”, “ $X + Y = Z$ ”

and

“Computer X is under attack by an intruder”

and

“Computer X is functioning properly”

Predicates

- These statements are neither true nor false
- When variables are substituted by values (elements) in the domain, the resulting statement is either true or false.
- The set of all such elements that make the predicate true is called the truth set of the predicate.

Predicates

- For example
 - Let $p(X)$ denote $X > 3$,
 - What are the truth values of $p(4)$ and $p(2)$?
 - $P(4)$, set $X = 4$,
 - $\Rightarrow 4 > 3$, **Which is true.**
 - $P(2)$, set $X = 2$,
 - $\Rightarrow 2 > 3$, **Which is false.**

Predicates

- For example
 - Let $Q(n)$ be the predicate “ n is a factor of 8”. Find the truth set of $Q(n)$ if,

- The domain of n is $\mathbb{Z}^+$

Truth set is $\{1, 2, 4, 8\}$

- The domain of n is $\mathbb{Z}$

Truth set is $\{1, 2, 4, 8, -1, -2, -4, -8\}$

The Universal Quantifier

- One sure way to change predicated into statements is to assign specific values to all their variables.
- Another way is to add quantifiers. Words that refer to quantities such as “some” or “all”
- The symbol $\forall$ denotes “for all” and is called the **universal quantifier**.
- Other expressions: For every, for arbitrary, for any, for each, given any.

The Universal Quantifier

- Let $Q(x)$ be a predicate and D the domain of x
- Universal statement is a statement of the form

$$\forall x \in D, Q(x)$$

- True if and only if $Q(x)$ is true for every x in D .
- False if and only if $Q(x)$ is false for at least one x in D
- Value of x for which $Q(x)$ is false is called Counterexample.

The Universal Quantifier

□ **Example:** Truth and falsity of Universal Statement.

- Let $D = \{1, 2, 3, 4, 5\}$, and consider the statement

$$\forall x \in D, x^2 \geq x$$

Show that the statement is true.

- Consider the statement $\forall x \in \mathbf{R}, x^2 \geq x$

Find a counterexample to show that this statement is false.

□ **Solution:** Check that “ $x^2 \geq x$ ” is true for each individual x in D .

$$1^2 \geq 1, 2^2 \geq 2, 3^2 \geq 3, 4^2 \geq 4, 5^2 \geq 5$$

Hence “ $\forall x \in D, x^2 \geq x$ ” is true.

Counterexample: Take $x = \frac{1}{2}$, then x is in $\mathbf{R}$ (since $\frac{1}{2}$ is a real number) and

$$\left(\frac{1}{2}\right)^2 = \frac{1}{4} \not\geq \frac{1}{2}$$

Hence “ $\forall x \in \mathbf{R}, x^2 \geq x$ ” is false.

The Existential Quantifier

□ The symbol $\exists$ denotes “There Exists”.

For Example: The sentence “There is a student in Math” can be written as

“ $\exists$ a person p such that p is a student in Math”

Or more Formally

“ $\exists p \in P$ such that p is student in math”

Other Expressions:

There is a , we can find a, there is at least one, for some, for at least one.

The Existential Quantifier

Let $Q(x)$ be the predicate, and D the domain of x .

An existential statement is a statement of the form,

$$\text{“}\exists p \in P \text{ such that } Q(x)\text{”}$$

True, if and only if, $Q(x)$ is true for at least one x in D .

False, if and only if, $Q(x)$ is false for all x in D .

The Existential Quantifier

□ **Example:** Truth and falsity of Existential Statement.

- Consider the statement

$$\exists m \in \mathbb{Z}^+, \text{ such that } m^2 = m$$

Show that the statement is true.

- Let $E = [5, 6, 7, 8]$ and consider the statement

$$\exists m \in E, \text{ such that } m^2 = m$$

Show that this statement is false.

□ **Solution:**

- Observe that $1^2 = 1$. Thus $m^2 = m$ is true for at least one integer m .
hence “ $\exists m \in \mathbb{Z}$ such that $m^2 = m$ ” is true.

- Note that $m^2 = m$ is not true for any integers m from 5 through 8:

$$5^2 = 25 \neq 5, 6^2 = 36 \neq 6, 7^2 = 49 \neq 7, 8^2 = 64 \neq 8$$

Hence “ $\exists m \in E, \text{ such that } m^2 = m$ ” is false.

Universal Conditional Statement

Probably, the most important form of statement

$\forall x$, if $P(x)$ then $Q(x)$.

□ Example:

- $\forall x \in \mathbf{R}$, if $x > 2$ then $x^2 > 4$

- Whenever a real number is greater than 2, its square is greater than 4.

OR

- The square of any real number greater than 2 is greater than 4.
- Translating sentences in English into logical expression is crucial task in mathematics, logic, programming, artificial intelligence, and many other disciplines.

Equivalent forms of Universal Statement

$\forall x \in U$, if $P(x)$ then $Q(x)$. Can always be written as

$$\forall x \in D, Q(x) \quad \left[\begin{array}{l} \text{By narrowing } U \text{ to } D \\ \text{consisting of all values of} \\ x \text{ that make } P(x) \text{ true} \end{array} \right.$$

□ Example:

- $\forall$ real numbers x , if x is an integer then x is rational. Means the same as
- $\forall$ integers x , x is rational.

$\forall x \in \mathbf{R}$ if $P(x)$ then $Q(x)$

$\forall x \in \mathbf{Z}, Q(x)$

$P(x) = x$ is an integer

$Q(x) = x$ is rational

Equivalent forms of Universal Statement

$\exists x$, such that $P(x)$ and $Q(x)$

Can always be written as

$\exists x \in D$, such that $Q(x)$

where D is the set of all x for which $P(x)$ is true.

Equivalent forms of Universal Statement

- **Example:** Express the proposition: “All human beings are intelligent” in predicate logic.

Let $I(x)$ = x , is intelligent

$H(x)$ = x is human being

Then

$$\forall x, H(x) \rightarrow I(x)$$

Or Equivalently

$$\forall x \in D, I(x)$$

Where D is a set of all x for which $H(x)$ is true.

Equivalent forms of Existential Statement

□ **Example:** Use the predicate logic to express the system specification.

1. Every mail message larger than 1 megabyte will be compressed.
2. If a user is active, at least one network link will be available.

□ **Solution:**

Let $S(m,y)$ = Mail message larger than Y megabyte

$S(m,1)$ = Mail message larger than 1 megabyte

$C(m)$ = Mail message will be compressed, Then

$$\forall m, S(m, 1) \rightarrow C(m)$$

- Second problem do it yourself.

Negation of a Universal Statement

$$\forall x \in D, Q(x)$$

Its negation is logically equivalent to

$$\exists x \in D, \neg Q(x)$$

That is

$$\neg(\forall x \in D, Q(x)) \equiv \exists x \in D \wedge \neg Q(x)$$

$\neg$ (all are) $\equiv$ “Some are not” or “There is at least one that is not”

Nested Quantifiers

- Quantifier within the scope of another quantifier

$\forall x \exists y (x + y = 0)$, where the domain of x and y consists of all real numbers.

- For every real number x there is a real number y such that, $x + y = 0$.

OR

- Every real number has an additive inverse.

□ Commonly occur in mathematics and computer science.

Nested Quantifiers

□ Example:

- Translate into English the following

$$\forall x, \forall y, ((x > 0) \wedge (y < 0) \rightarrow (xy < 0))$$

where the domain of x and y consists of all real numbers.

Nested Quantification as Loops

- It is sometimes easier to think of nested quantification in terms of nested loops.

$$\forall x, \forall y, P(x, y)$$

- To see whether it is true.
 - Loop through the values of x
 - For each x , loop through the values of y
 - If $P(x, y)$ is true for all values of x and y , (o) is true.
 - If we ever hit a value x , for which we hit a value y for which $P(x, y)$ is false, (o) is false.

Nested Quantification as Loops

$$\forall x, \exists y, P(x, y)$$

- To see whether it is true.
 - Loop through the values of x
 - For each x , loop through the values of y , until we find a y for which $P(x, y)$ is true.
 - If for every x , we hit such a y , (o) is true
 - If for some x , we never hit such a y , (o) is false.

Nested Quantification as Loops

$$\exists x, \forall y, P(x, y)$$

$$\exists x, \exists y, P(x, y)$$

“Try it Yourself”

Negating Nested Quantification

□ Statements involving nested quantifiers can be negated by successively applying rules of negating single quantifier statements.

□ For Example

- Express the negation of $\forall x, \exists y (xy = 1)$
- $\sim(\forall x, \exists y (xy = 1)) \equiv \exists x \sim \exists y (xy = 1)$
- $\sim(\forall x, \exists y (xy = 1)) \equiv \exists x \forall y (xy = 1)$
- $\sim(\forall x, \exists y (xy = 1)) \equiv \exists x \forall y (xy \neq 1)$

Arguments with Quantified Statements

□ Universal modus ponens.

$\forall x$, if $P(x)$ then $Q(x)$

$P(a)$ for a particular a

 $\therefore Q(a)$

$\forall x (p(x) \rightarrow Q(x))$

$P(a)$ for a particular a

 $\therefore Q(a)$

Different Representations of
the same concept (argument
form)

Arguments with Quantified Statements

❑ **Example:** Is this argument valid? Why?

- If an integer is even, then its square is even
k is a particular integer that is even
 $\therefore k^2$ is even

❑ **Solution:** Let $E(x)$ be “an integer is even”

- $S(x)$ be “its square is even”, Then

$\forall x, \text{ if } E(x) \text{ then } S(x)$

$E(k)$ for a particular k

 $\therefore S(k)$

- This argument has the form of “universal modus ponens”. Therefore it is a valid argument.

Arguments with Quantified Statements

□ Universal modus Tollens:

$\forall x$, if $P(x)$ then $Q(x)$

$\sim Q(a)$ for a particular a

 $\therefore \sim P(a)$

$\forall x, \forall x (p(x) \rightarrow Q(x))$

$\sim Q(x)$ for a particular a

 $\therefore \sim P(a)$

Arguments with Quantified Statements

- ❑ **Example:** Recognizing the form of Universal Modus Tollens.
- Rewrite the following argument using quantifiers, variables, and predicate symbols. Write the major premise in conditional form. In this argument valid? why?

All Human beings are mortal

Zeus is not mortal.

 $\therefore$ Zeus is not human

- ❑ **Solution:** The major premise can be rewritten as $\forall x$, if x is human then x is mortal.

Let $H(x)$ be “ x is human”, let $M(x)$ be “ x is mortal”, and let Z stand for Zeus. The argument becomes.

$\forall x$, if $H(x)$ then $M(x)$

$\sim M(Z)$

 $\therefore \sim M(Z)$

This argument has the form of universal modus Tollens and is therefore valid.

Proof

Proofs

□ Simply “A Mathematical Proof is a carefully reasoned argument to convince a skeptical listener.

□ **Importance:**

- If you have a conjecture, the only way you can safely be sure about its correctness is by presenting a valid proof.
- While trying to prove something, we may gain a great deal of understanding and knowledge, even if we fail.

Proofs

□ More formally

- “A mathematical proof of a proposition is a chain of logical deductions leading to the proposition from a bas set of axioms.

□ Proposition:

□ Logic:

□ Axioms:

- Propositions that are simply accepted as true.
- For Example: “There is a straight line segment between every pair of points”.

Proofs

□ Proving Implications:

- Many theorems are of the form
 - $P \Rightarrow Q$, P implies Q, or If P then Q.
- If $ax^2 + bx + c = 0$ and $a \neq 0$, then
 - $$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
- If n is an even integer greater than 2, then n is a sum of two primes.

□ To prove: Method 1:

- Write , “Assume P”
- Show that Q logically follows

Proofs

□ Proving Implications:

□ Example

For any two positive numbers x and y ,

$$\bullet \quad \sqrt{xy} \leq \frac{x+y}{2}$$

Proof: Assume x and y are positive numbers,

$$\bullet \quad (\sqrt{x} - \sqrt{y})^2 \geq 0$$

By algebra this implies

$$\bullet \quad x + y - 2\sqrt{x}\sqrt{y} \geq 0$$

Moving $2\sqrt{x}\sqrt{y}$ to the other side, we get that.

$$\bullet \quad x + y \geq 2\sqrt{x}\sqrt{y}$$

Dividing both sides by 2 yields the inequality

$$\bullet \quad \frac{x+y}{2} \geq \sqrt{xy} \text{ or } \sqrt{xy} \leq \frac{x+y}{2}$$

Proofs

□ Proving Implications:

- Method 2: Prove the contrapositive.
 - $\sim P \Rightarrow \sim Q$
 - $P \Rightarrow Q \equiv \sim P \Rightarrow \sim Q$

Therefore, proving one is as good as proving the other. Proceed as follows,

- Write “We prove the contrapositive.” and then state the contrapositive.
- Proceed as in Method #1.

Proofs

Example:

If r is irrational, then $\sqrt{r}$ is also irrational

Proof:

We prove the contrapositive: If $\sqrt{r}$ is rational, then r is rational.

Assume $\sqrt{r}$ is rational. Then there exist integers m and n such that,

$$\sqrt{r} = \frac{m}{n}$$

Squaring both sides

$$r = \frac{m^2}{n^2}$$

Since m^2 and n^2 are integers, so r is rational.

Proofs

□ **Direct Proof:**

Method 1 is an example of “Direct Proof”.

Lets look at some more examples of Direct proof.

Proofs

□ **Example:** For all integers a and b , if a and b are positive and a divides b , then $a \leq b$.

Proof: Assume a and b are positive integers and a divides b .

Then there exists an integer k such that,

$$B = a * k$$

Since both a and b are positive, it follows that

$$1 \leq k$$

Multiplying both sides by a we get

$$a \leq a * k = b$$

$$a \leq b$$

Proofs

□ **Example:** For all integers a , b , and c , if a divides b and b divides c , then a divides c .

Proof: Assume a , b , and c are integers, such that a divides b , and b divides c .

Then there exist integer r and s , such that,

$$b = a*r \text{ and } c = b*s$$

By substitution

$$c = (a*r)*s$$

$$c = a*(r*s)$$

Let $k=r*s$, then k is an integer, since it is a product of integers, and therefore

$$c = a*k$$

Thus a divides c .

Proofs

Proof by Division into cases.

- Breaking a proof into cases, and proving each case separately.
- Useful for complicated proofs.

□ **Example:** Any two consecutive integers have opposite parity.

Proof: Assume m and $m+1$ be the two consecutive integers. We break the proof in two cases.

Case 1: (m is Even): This means $m=2k$, for some integer k . Thus

$$m+1 = 2k + 1$$

which is odd, by definition of odd numbers. Hence in this case one of “ m ” and “ $m+1$ ” is even and the other is odd.

Proofs

Proof by Division into cases.

Proof: Case 2: (m is Odd): in this case,

$$m = 2k + 1, \text{ for some integer } k$$

Thus

$$\begin{aligned} m + 1 &= 2k + 1 + 1 \\ &= 2k + 2 \\ &= 2(k+1) \end{aligned}$$

But $k+1$ is integer, as it is a sum of two integers.

Thus $m+1$ is twice some integers, hence $m+1$ is even.

Hence in this case also one of “ m ” and “ $m+1$ ” is even and the other is odd.

Proofs

Proof by Division into cases.

- This is another type of “Direct Proof”
- Lets look at another example.

□ **Example:** “If a and b are any integers not both zero, and if q and r are any integers such that.

$$a = b*q + r$$

Then

$$\gcd(a,b) = \gcd(b,r)$$

(gcd= Greatest Common Divisor)

Proofs

Indirect Proofs:

- Proof by contrapositive is an example of indirect proof.
- Another common kind of indirect proof is “proof by contradiction”.
- “You show that if a proposition were false, then same false fact would be true.”

□ Method:

1. Write, “We use proof by contradiction”
2. Write, “Suppose P is false”
3. Deduce something known to be false (Logical contradiction).
4. Write “This is a contradiction, therefore P must be false.”

Proofs

Proof by Contradiction:

- **Example:** $\sqrt{2}$ is Irrational.
- **Proof:** We use proof by contradiction. Suppose $\sqrt{2}$ is rational. Then there exist integers m and n with no common factors such that.

$$\sqrt{2} = \frac{m}{n}$$

Squaring both sides gives

$$2 = \frac{m^2}{n^2} \Rightarrow m^2 = 2n^2$$

This means that m^2 is even, which implies that m is even. That is for some integer k .

$$\begin{aligned} m &= 2k \\ m^2 &= (2k)^2 = 4k^2 = 2n^2 \end{aligned}$$

Or equivalently

$$n^2 = 2k^2$$

Consequently n^2 is even, and so n is even.

Hence both “ m ” and “ n ” have a common factor of 2, which contradicts our supposition. Hence the supposition is false, and the theorem is true.

Proofs

Proof by Contradiction:

- ❑ **Example:** $\sqrt{2}$ is Irrational using Geometry.
- ❑ **Proof:** Uses the idea of Commensurability.
(Ch:1, P:8-10)

Proofs

Proof by Contradiction:

Example:

- ❑ **Theorem:** “The Quadrature of a circle is impossible”.

- ❑ **Proof:** Uses the concept of Algebraic numbers, Constructible numbers, and transcendental numbers. (Ch:1, P:25)

Proofs

Proof by Contradiction:

Example

□ **Theorem:** “If a straight line falling on two straight lines makes the alternate angles equal to one another, the straight lines will be parallel”.

□ **Proof:** Ch:2, P:44.

Proofs

Summary:

- ☐ **Direct Proof**
- ☐ **Disproof by Counter example**
- ☐ **Proof by Contradiction**
- ☐ **Proof by Contrapositive**